Sai Priya Palla

Senior Generative AI Engineer| Machine Learning Engineer|Data Scientist
saipriyakrr@gmail.com | +1(214)9370262

PROFESSIONAL SUMMARY:

Senior AI/ML Engineer/ Data Scientist with 7+ years of experience delivering production-ready machine learning and LLM solutions across regulated domains (banking/financial services) and healthcare. Proven expertise in building LLM assistants and Retrieval-Augmented Generation (RAG) systems using GPT-4, Claude, Llama 2, LangChain, Pinecone, and ChromaDB, with strong governance controls including citation-backed responses, auditable traces, access-aware retrieval, and safety guardrails (PII masking, sensitive-intent handling, prompt-injection defenses). Strong background in predictive modeling, time-series analytics, NLP, and computer vision using LightGBM, XGBoost, Random Forest, LSTM, BERT/BioBERT, and YOLO. Experienced in end-to-end ML lifecycle delivery, including data pipelines (Airflow), API services (FastAPI/Flask), and scalable deployment on AWS (SageMaker, EKS/EC2) with Docker/Kubernetes, monitoring via MLflow, and performance optimization. Adept at partnering with risk, compliance, clinical, and product stakeholders to translate requirements into secure, reliable, and measurable outcomes.

Core Expertise:

- **Generative AI & LLMs:** GPT, BERT, Hugging Face Transformers, fine-tuning and deployment of foundation models, conversational AI, intelligent search, and document retrieval.
- **MLOps & Cloud Infrastructure:** LLMOps, Vertex AI, CI/CD, Docker, Kubernetes, Jenkins, Terraform, MLflow, Airflow, AWS SageMaker, BigQuery, Azure ML, multi-cloud ML pipelines.
- **Fraud Detection & Risk Analytics:** Real-time anomaly detection, risk scoring, compliance monitoring, transaction security, and adaptive fraud models.
- **Big Data & Streaming Analytics**: Kafka, Spark, PySpark, Kinesis, Databricks, large-scale ETL pipelines, feature engineering, and real-time analytics.
- **Statistical Modeling & Data Analytics:** Proficient in regression, classification, hypothesis testing, and multivariate analysis using Python (NumPy, SciPy, Scikit-learn) for large-scale financial and transactional datasets.
- **Leadership & Strategy**: Cross-functional collaboration, enterprise AI adoption, team mentoring, and executive-level stakeholder engagement.
- **Observability & Evaluation for LLMs**: Langfuse (rapid adoption), MLflow, tracing/latency/error analysis, prompt/version tracking, offline/online evals.
- **Document Intelligence & RAG:** RESTful APIs, GraphQL, FastAPI, Flask, API versioning, schema-driven responses.
- **Datastores: PostgreSQL**, **MongoDB**, Redis (caching), indexing & query optimization

Technical Proficiencies:

Programming & Frameworks:
- Python (TensorFlow, PyTorch, Keras, Hugging Face, Scikit-learn, Pandas, NumPy)
- Scala (Spark, Spark SQL, PySpark)
- Golang (REST API & Microservices)
- SQL (PostgreSQL, MySQL, Redshift)
- Kotlin (API Workflows, JVM Ecosystem Exposure)

Machine Learning & Deep Learning:
- Generative AI, LLMs, Transformers (GPT, BERT)
- CNNs, RNNs, GNNs, Recommendation Systems, Anomaly Detection
- Predictive Analytics, Time-Series Forecasting, Sentiment Analysis
- Computer Vision (OpenCV, Image/Video Analytics, Camera-based Monitoring)

Natural Language Processing:
- Text Classification, Named Entity Recognition (NER), Topic Modeling, LLM-based systems
- LLM-based Systems, Compliance Monitoring, Document Intelligence (RAG, evaluators), LlamaParse.

Analytical & Statistical Techniques:
- Regression, Classification, Time-Series Forecasting, A/B Testing, ANOVA, PCA, Clustering, and Hypothesis Testing.

Backend & Distributed Systems:
- REST & Async Microservices
- Event-Driven Architectures & Messaging
- JVM Ecosystem & Kotlin (API workflows / ML service integration)
- API Design, RBAC, Authentication, Multi-tenant Security
- Distributed Systems Design & Service Orchestration

MLOps & DevOps:
- Docker, Kubernetes, Jenkins, MLflow, Airflow
- CI/CD Pipelines, Model Monitoring, Drift Detection & Retraining, LLM Observability: Langfuse
- Infrastructure Automation: Terraform, Ansible

Orchestration platforms:
- Kubernetes, Airflow, SageMaker Pipelines
- Argo, event-driven microservices
- Distributed workflow design

Graph Algorithms & Systems:
- Strong DSA Fundamentals
- BFS / DFS for Graph Traversal & Dependency Flows
- Computational Graph Execution (Model DAGs / Pipeline DAGs)
- Network Graph Analytics (Fraud signal propagation / anomaly mapping)

Cloud & Platforms:
- AWS (S3, SageMaker, Redshift, Lambda, Glue)
- GCP (BigQuery, Dataproc, Vertex AI)
- Azure ML (Azure AI Studio, Cognitive Search, Azure ML, Microsoft Fabric, Power BI, Copilot Studio, Microsoft Power Platform)
- Multi-cloud ML Pipelines (LLMOps, ModelOps, MLOps best practices)

Big Data & Data Engineering:
- Hadoop, Spark, Spark Streaming, Kafka, Kinesis, Databricks
- AWS Glue, Lambda, Feature Engineering, Large-scale ETL Pipelines

Visualization:
- Tableau, Grafana, Power BI, Microsoft Fabric

Version Control / Project Management:
- Git, JIRA, Confluence

EDUCATION:

Sreyas Institute of Engineering and Technology, India, *B.Tech, Computer Science | Aug 2014 – May 2018*

PROFESSIONAL EXPERIENCE:

Senior AI Engineer/ML Engineer | Fifth Third Bank, Cincinnati, OH

March 2025 – Present

Environment: *Python, PyTorch, LangChain, GPT-4, Claude, Llama 2, Pinecone, ChromaDB, FastAPI, AWS SageMaker, Docker, Kubernetes,*

PostgreSql, Redis

- Led end-to-end delivery of an internal banking LLM assistant (GPT-4 + LangChain) for account/loan/transaction servicing workflows, producing grounded, policy-aligned responses with clear handoff paths for unsupported or sensitive requests.
- Architected a production RAG platform (Pinecone/ChromaDB) indexing **300–1,500** internal policy/compliance/product documents (**10k–20k** chunks); improved retrieval relevance by **10–25%** after chunking strategy + metadata tagging + reranking, reducing hallucination-prone responses.
- Built an auditable "answer-with-sources" experience: **90–98%** of responses include citations; achieved **85–95%** citation correctness in sampled reviews by enforcing citation-required output schemas and traceable retrieval context.
- Implemented long-document Q&A for mortgage/credit workflows using Claude; supported **20–80** long-context cases/week with structured extraction outputs; raised downstream extraction consistency to **85–95%** schema-valid responses using validation rules.
- Fine-tuned Llama 2 with LoRA on domain terminology and approved Q&A patterns; introduced model routing (frontier vs hosted) to balance quality, latency, and cost while keeping outputs within banking policy boundaries.
- Developed a secure document intelligence pipeline for PDFs/scanned forms using OCR + NER to extract key entities (names, dates, amounts, addresses, identifiers), with confidence scoring and human-review fallback for low-confidence cases.
- Engineered FastAPI microservices for retrieval + LLM orchestration with Redis caching, rate limiting, request timeouts, idempotent retries, and structured logging—built for reliability and operational scale.
- Designed and delivered RESTful APIs in FastAPI for RAG + LLM orchestration (retrieval, ranking, citations, guardrails), using versioned endpoints, pagination, and consistent response schemas.
- Built fraud and risk alerting capabilities by combining classical ML (XGBoost) with transformer-based representations for transactional patterns; produced analyst-facing signals and explanations designed for triage—not automated final decisions.
- Implemented sentiment + topic analytics over customer feedback and chat transcripts using fine-tuned BERT-style models to identify recurring drivers of dissatisfaction, escalation themes, and product/service gaps.
- Deployed LangChain services behind RESTful APIs (FastAPI) and persisted run metadata/citations in PostgreSQL for audit and analytics.
- Established modern LLM evaluation (beyond ROUGE/BLEU): retrieval precision/recall@k, groundedness/faithfulness checks, citation accuracy, refusal correctness, and safety-policy compliance; created regression suites for prompt/model releases.
- Set up MLOps for model packaging and deployment on AWS EKS/SageMaker: Dockerized builds, environment-based configuration, staged rollout practices, and operational runbooks for supportability.
- Implemented monitoring via MLflow + dashboards tracking latency, error rate, cost/request, cache hit rate, and quality signals; enabled rapid rollback and version control across models, prompts, and retrieval configs.
- Optimized inference via batching/caching and INT8 quantization where suitable; reduced compute cost by **15–35%** while maintaining quality thresholds and safety constraints.
- Integrated explainability (SHAP) for credit/risk decision-support models to make key drivers interpretable for stakeholders and compliance review, supporting audit readiness and transparent reporting.

- Collaborated with risk, compliance, product, and operations teams to define requirements, approval workflows, and documentation standards; ensured solutions met internal controls, security expectations, and regulated banking practices.

AI Engineer II | Davita, Hyderabad, India

2020 – 2024

Environment: *Python, TensorFlow, PyTorch, BioBERT, spaCy, YOLO, ResNet, Flask, AWS (SageMaker, EC2, S3), Docker, Airflow,*

MongoDB, Postgresql

- Developed and deployed an active readmission risk prediction model using LightGBM, translating dialysis patient history into actionable risk scores for care-team prioritization.
- Built a production vascular access failure prediction model using Random Forest, leveraging longitudinal treatment history and clinical trends to surface high-risk patients early.
- Implemented anomaly detection on vitals/labs (Isolation Forest + LSTM) monitoring **10k–50k** patient-days/month; reduced alert noise by **20–45%** after threshold tuning with clinical stakeholders.
- Designed time-series feature engineering for dialysis populations (rolling windows, slopes/trends, recent deltas, volatility, missingness patterns) across vitals, labs, and medication/treatment history to support both tabular and sequence models.
- Built a clinical NER pipeline using BioBERT to extract diagnoses, medications, and lab values from unstructured notes; added normalization for common abbreviations/terminology to improve consistency for downstream use.
- Created clinical note type classification to automatically categorize documentation (e.g., discharge summaries, progress notes), improving routing and making NLP coverage more reliable across note sources.
- Built LangChain-based RAG pipelines (loaders → chunking → embeddings → retrievers) to answer user queries using internal documents with grounded responses.
- Built an Apache Airflow data pipeline to ingest and refresh patient data from EHR and lab systems; implemented data validation checks and backfill handling to keep features reliable for active models.
- Implemented GraphQL (where needed) to aggregate multi-source responses (policy docs, metadata, evaluation signals) into a single query for UI/client teams, reducing client-side calls and simplifying integration.
- Delivered predictions through a Flask REST API consumed by EMR-connected services and clinical dashboards, using stable response schemas and guardrails to ensure consistent behavior under real clinical usage.
- Deployed and managed model endpoints on AWS SageMaker with autoscaling and version control; introduced staged releases and safe rollback practices for production updates.
- Implemented automated deployment workflows using Jenkins + Docker, standardizing packaging, testing, and promotion of model services across environments.
- Set up model monitoring with MLflow, tracking prediction stability, drift signals, and operational metrics (latency, errors) to trigger investigation/retraining when performance degraded.
- Implemented privacy/security controls aligned with healthcare expectations, including encryption, strict access control, and audit logging for systems handling patient data and model outputs.
- Collaborated with clinicians and data engineers to validate features, review error cases, and embed models into workflows as decision support (clinician-in-the-loop), ensuring safe adoption and operational fit.

AI Engineer | ServiceNow, Hyderabad, India

2018 – 2020

Environment: *Python, TensorFlow, LSTM, BERT, spaCy, NLTK, Flask, AWS EC2, Redis, PostgreSQL, Git*

- Built IT helpdesk chatbot supporting **8–20** common intents (password reset, software request, access issues), deflecting **10–30%** of routine L1 tickets in pilot usage.
- Built ticket auto-routing for L1/L2/L3 queues; reduced manual triage time by **20–40%** and lowered misroutes by **10–25%** after rule + model refinement.
- Implemented entity extraction using spaCy identifying usernames, systems, error codes from tickets
- Created ticket auto-routing system using text classification to assign L1/L2/L3 support teams
- Built sentiment analysis on ticket descriptions to prioritize urgent issues
- Developed ticket similarity search using TF-IDF, cosine similarity suggesting previous solutions
- Implemented knowledge base search using semantic search to recommend articles
- Created Flask API for chatbot integration with ServiceNow platform
- Built data preprocessing pipeline for text cleaning, tokenization, lemmatization
- Deployed models on AWS EC2 with Redis caching for fast responses
- Implemented model evaluation using accuracy, F1-score, confusion matrix
- Collaborated with IT support, DevOps teams for chatbot deployment, maintenance

Certifications

- Microsoft Certified: Azure AI Engineer Associate (AI-102)
- Databricks Certified Generative AI Engineer Associate